

UniSense – Definitive Product & Growth Document

Product Name: UniSense

Category: AI-Powered University Operating System (SaaS + API)

Primary Market: Africa-first, Global-ready

Stage: MVP → Global Infrastructure Platform

1. What UniSense Is (In Simple Terms)

UniSense is a full operating system for universities.

It replaces fragmented tools (Excel, paper, WhatsApp groups, legacy SIS/LMS software) with **one intelligent, unified platform** that manages: - Students - Courses & exams - Fees & finance - Communication - Analytics & AI insights

UniSense is **not just software** — it is **education infrastructure**.

2. Core Problem UniSense Solves

Universities today face: - Disconnected systems (SIS ≠ Finance ≠ Communication) - Manual, error-prone workflows - No predictive insight (dropouts, GPA risk, course demand) - Poor UX for students and staff - Systems not designed for low bandwidth or mobile usage

These problems cost universities **money, time, and student success**.

3. UniSense Solution Overview

UniSense provides: - A **single source of truth** for university data - Role-based dashboards for every stakeholder - Built-in AI for decision-making - Offline + mobile + WhatsApp-first communication - Modular SaaS pricing (start small, expand)

4. Target Users

- University Admin / Registrar
- Faculty Deans / HODs
- Lecturers / Professors
- Students
- Finance / Bursary Officers

Each user has a **personalized dashboard** with only relevant actions and insights.

5. Core Product Modules (MVP)

5.1 User & Access Management

- Email/password authentication
- Role-based permissions
- Multi-tenant university isolation

5.2 Student Information System (SIS)

- Student profiles
- Enrollment & academic status
- Faculty & department mapping
- Bulk CSV import

5.3 Course & Credit Management

- Course creation & credit units
- Lecturer assignment
- Student course registration
- Credit limit enforcement

5.4 Exams, Grading & GPA

- Continuous assessment & exams
- Automated grading
- GPA & CGPA calculation
- Results publishing

5.5 Fees & Finance (MVP Scope)

- Fee structures
- Student invoices
- Payment status tracking (manual)
- Financial reports

5.6 Communication

- Announcements
 - Email & SMS notifications
 - WhatsApp notifications (via FastAPI microservice)
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6. AI & Analytics (Included in MVP)

At-Risk Student Detection

- Flags students likely to fail or drop out
- Inputs: GPA trends, failed courses, fee status

GPA Prediction

- Predicts end-of-semester GPA
- Early warning indicators

Lecturer Workload Analytics

- Teaching load vs student count
- Overload detection

Admin Insights

- Enrollment trends
- Pass/fail heatmaps
- Department performance

AI starts **lightweight** and becomes more powerful as data grows.

7. Technical Philosophy (High Level)

- Web-based SaaS (React + Node.js)
 - PostgreSQL (relational, reliable)
 - Python-based AI services
 - WhatsApp handled via **separate FastAPI microservice**
 - API-first, modular, scalable
 - Offline-friendly (PWA)
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8. Business Model

Primary Revenue

- SaaS pricing per student per year

Secondary Revenue

- Paid add-on modules
- API access (data, AI, compliance)
- Enterprise & government contracts

9. Phase-by-Phase Growth Plan

◆ Phase 0: MVP + Validation (0–12 Months)

Goal: Prove universities will use and pay for UniSense.

Features

- Core academic modules
- Fees tracking
- Role-based dashboards
- WhatsApp communication
- Basic AI analytics

Customers

- 3–5 private universities

Revenue (Conservative)

- ~15,000 students
- ~\$15k ARR

Outcome

- Product-market validation
 - Real university data
 - Case studies
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◆ Phase 1: Product-Market Fit (Year 1–2)

Goal: Become mission-critical software.

Added Features

- Online payments
- Timetable automation
- Dropout prediction
- Smart notifications

Customers

- ~25 universities

Revenue

- ~\$200k+ ARR

Outcome

- Strong retention
 - Expansion within universities
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◆ Phase 2: Regional Scale (Year 2–4)

Goal: Become education infrastructure across Africa.

Added Features

- Hostel & accommodation
- Research & grants tracking
- Multi-campus support
- Advanced AI forecasting

Customers

- ~150 universities

Revenue

- ~\$2–3M ARR

Outcome

- Regional dominance
 - API partnerships
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◆ Phase 3: Global Platform (Year 4–6)

Goal: Become a global University OS.

Added Features

- International compliance
- Alumni & career tracking
- Global research collaboration AI
- Marketplace & ecosystem

Customers

- 500+ universities

Revenue

- \$30M+ ARR

Outcome

- Infrastructure-level company
 - Unicorn-scale valuation
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10. Competitive Positioning

UniSense is positioned as:

“The Operating System for Universities in Emerging Markets and Beyond.”

Key Differentiators

- AI-native from day one
 - Offline + WhatsApp-first
 - Modular & scalable
 - High switching costs
 - Valuable long-term data moat
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11. Long-Term Vision

UniSense becomes: - The default backend for universities - The data layer for education policy - The API platform powering EdTech innovation

12. Final Statement

UniSense is not a feature product. It is **foundational infrastructure** for higher education.

This document serves as the **canonical reference** for building, scaling, pitching, and operating UniSense.